MAVUNO HOME & CARE

Revenue Decline Diagnostic

Why group revenue fell 10.7% — and who owns each cause

Comparison period	Q3 2026 vs Q3 2025
Prepared	2026-10-05
Prepared by	Senior Data Analyst
Classification	Internal — Commercial leadership
Reproduce with	python3 src/run_pipeline.py

The question this answers

Whose plan gets rewritten next quarter — retention, pricing, or merchandising?

Group net revenue fell KSh 5.21M (10.7%). We did not lose customers — active customers rose +5.5%. We lost trips (-11.4% orders per customer) and basket (-4.5% average order value), and 71% of the damage sits in two of six stores.

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SECTION 1

Executive summary

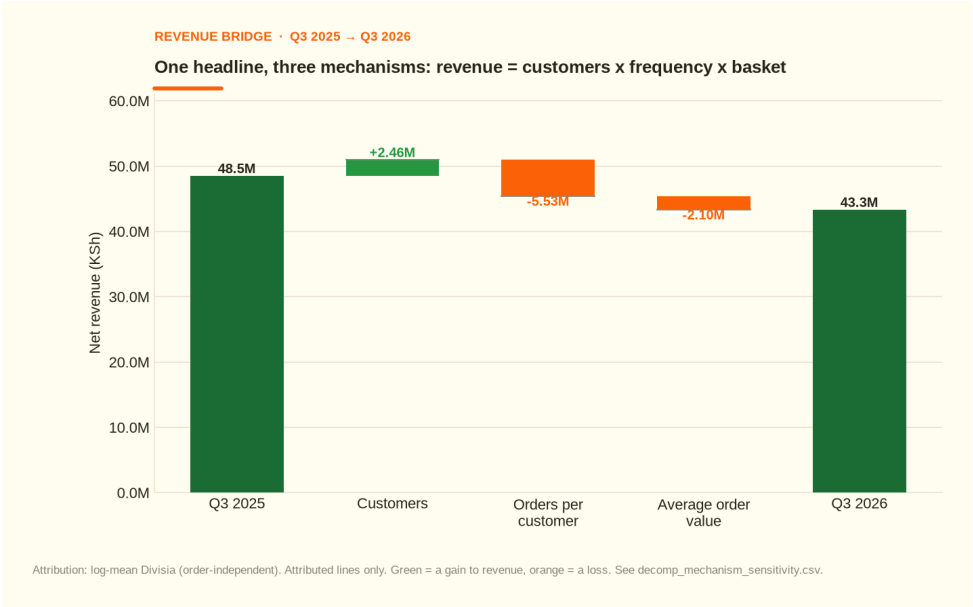
NET REVENUE	GROSS MARGIN	ORDERS PER CUSTOMER
KSh 43.65M	**18.57%**	**4.82**
-10.7% from KSh 48.86M in Q3 2025	**-1.38pp** gross profit KSh 8.04M	**-11.4%** the engine of the decline
AVERAGE ORDER VALUE	**ACTIVE CUSTOMERS**	**REVENUE WITH NO CUSTOMER ID**
KSh 1,619	**5,555**	**0.76%**
-4.5% 3.40 lines per order	**+5.5%** acquisition is NOT the problem	**KSh 0.70M** the declared coverage limit

Group net revenue fell KSh 5.21M (10.7%). We did not lose customers — active customers rose +5.5%. We lost trips (-11.4% orders per customer) and basket (-4.5% average order value), and 71% of the damage sits in two of six stores.

Gross profit fell 16.9% against a 10.7% revenue fall, because gross margin compressed 1.38pp to 18.57%. Revenue is not profit: every recommendation below is therefore priced in gross profit, not revenue.

The four findings, ranked by how much of the decline they own

#	Owner	The problem	Cost	Share
1	Supply Chain & Merchandising	Two hero Personal Care SKUs out of stock for six weeks at Mombasa Nyali; the damage spread to whole baskets	KSh -2.16M	41%
2	Retail Operations (Thika Road)	Thika Road footfall cut for the full quarter — visits per customer well below every other store	KSh -1.49M	29%
3	CRM / Loyalty & Assortment	Residual: fewer trips per customer and slimmer baskets right across the chain	KSh -0.97M	19%
4	Pricing & Category Management	Beverage promotional depth deepened sharply outside the two affected stores	KSh -0.55M	11%



The 10.7% fall is one customer gain against two larger losses — it is not one event.

Figure 1 — the revenue bridge: one gain, two larger losses.

SECTION 2

The decision this supports

The question

Whose plan gets rewritten next quarter — retention, pricing, or merchandising?

An analysis is only worth reading to the extent it changes who does what on Monday. This one converts a single headline — revenue fell — into four named owners with a quantified problem, an expected recovery and a review date. If it does not change someone's plan next quarter, it has failed.

What this report is, and what a dashboard is

	This report	The dashboard that follows it
Job	explain a change and support a decision	monitor the KPIs the analysis identified
Question	“why, and what should we do?”	“are we OK?”
Cadence	one-off, on demand	monthly, always-on
Output	findings + recommendations, with owners	tiles, trends, exceptions

The dashboard ships after the thinking and monitors what this report concluded. Its build specification, including the DAX measures and the targets below, is docs/07_dashboard_spec.md.

SECTION 3

What the data is, and what it is not

Line-item grain: one row = one line item on one customer receipt — 195,224 lines across 55,431 invoices in the two quarters, plus a separate branch x SKU x day shelf-availability table that makes the supply event provable rather than plausible.

Declared limits — limits on the conclusions, not footnotes

- 0.76% of revenue carries no customer identifier (KSh 0.70M of anonymous walk-ins). Every segment-level conclusion therefore reconciles to 99.24% of revenue, not 100%. The reconciliation is shown as its own line rather than hidden in an 'Other' bucket.
- The dataset is simulated from a documented mechanism (docs/00_ground_truth.md). That is deliberate: it lets the diagnosis be graded. See outputs/validation_report.md — 14/14 checks pass against the sealed answer key.
- No procurement, lead-time or supplier-performance data exists in this extract. The fifth why therefore stops at a data-capture recommendation instead of a guess.
- The annualised figure is a stated run-rate (KSh -20.83M, the quarter's change x 4), not a forecast. It assumes the Mombasa Nyali incident does not repeat and ignores Q4 seasonality.

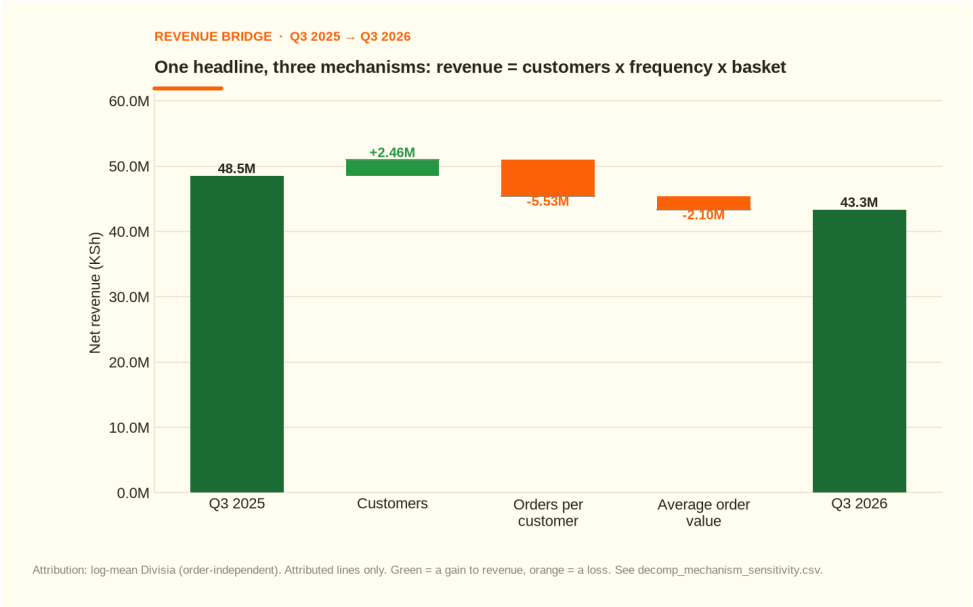
Method in one line: decompose by mechanism first (customers x frequency x basket), then by structure (store, category, segment, channel), then intersect the three, then ask why five times with a number attached to every answer. The attribution is log-mean Divisia, audited so the driver partition covers the decline exactly.

SECTION 4

Descriptive — what happened

Metric	Q3 2025	Q3 2026	Change
Net revenue	KSh 48.86M	KSh 43.65M	-10.7%
Gross profit	KSh 9.68M	KSh 8.04M	-16.9%
Gross margin	19.95%	18.57%	-1.38pp
Active customers	5,265	5,555	+5.5%
Orders	28,618	26,764	-6.5%
Orders per customer	5.44	4.82	-11.4%
Average order value	KSh 1,694	KSh 1,619	-4.5%
Lines per order	3.59	3.40	-5.2%
List price per unit	KSh 312	KSh 326	+4.3%
Discount depth (% of gross)	4.80%	7.82%	+3.02pp

Every figure in this table is reproduced from outputs/kpis.json, which the pipeline regenerates. Nothing here is typed by hand.



One customer gain against two larger losses. This is not one event.

Figure 1 — the revenue bridge.

SECTION 5

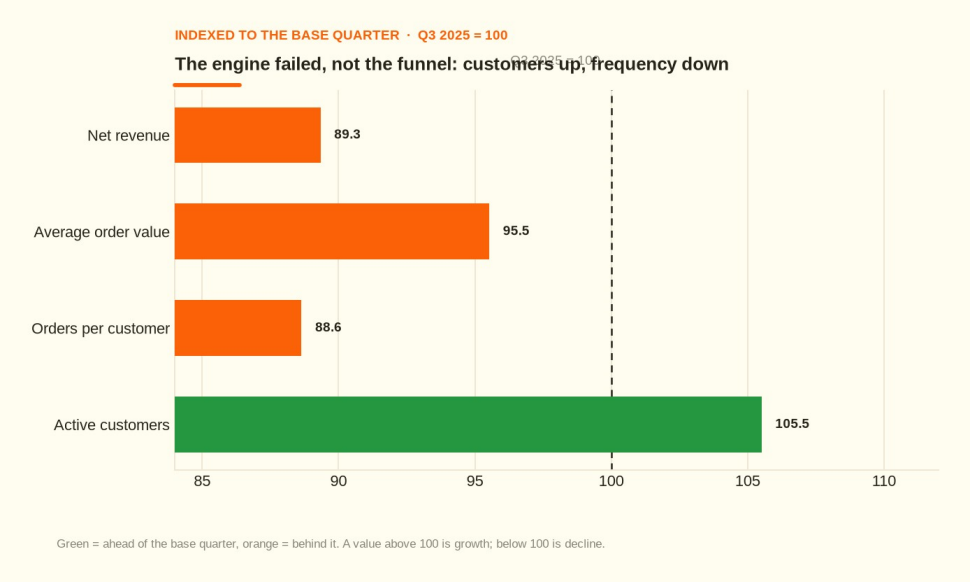
Diagnostic, part 1 — the mechanism

Revenue = Customers x Frequency x Average order value.

Driver	Effect on revenue	YoY move
Customers	KSh +2.46M	+5.5%
Orders per customer	KSh -5.53M	-11.4%
Average order value	KSh -2.10M	-4.5%
Total	KSh -5.21M	-10.7%

Attribution is log-mean Divisia, so the ranking does not depend on the order in which the three factors are multiplied. The frequency effect is the largest under every ordering — that is the robust part of the result.

Sequential-order sensitivity band: customers KSh 2.26M to KSh 2.67M; frequency KSh -5.81M to KSh -5.26M; aov KSh -2.17M to KSh -2.03M.



Customers grew. Orders per customer and basket value did not. Any report that stops at “fewer customers” is wrong.

Figure 2 — the same three drivers indexed to the base quarter.

SECTION 6

Diagnostic, part 2 — the structure, and who owns what

By store

Rank	Slice	Revenue change	YoY	Share of group decline
1	Mombasa Nyali	KSh -2.18M	-27.0%	41.9%
2	Thika Road	KSh -1.50M	-18.7%	28.9%
3	Kisumu Lakefront	KSh -0.69M	-11.5%	13.2%
4	Town Centre	KSh -0.39M	-3.4%	7.4%
5	Westlands	KSh -0.26M	-2.8%	5.0%
6	Nakuru CBD	KSh -0.19M	-3.0%	3.6%

By category

Rank	Slice	Revenue change	YoY	Share of group decline
1	Groceries	KSh -1.60M	-13.1%	30.7%
2	Beverages	KSh -1.08M	-20.3%	20.7%
3	Household & Cleaning	KSh -0.82M	-11.3%	15.7%
4	Personal Care	KSh -0.65M	-7.9%	12.4%
5	Baby & Kids	KSh -0.46M	-6.7%	8.8%
6	Dairy & Eggs	KSh -0.37M	-8.0%	7.0%
7	Fresh Produce	KSh -0.15M	-4.7%	2.9%
8	Bakery	KSh -0.09M	-6.9%	1.7%
9	Unclassified	KSh -0.00M	-27.9%	0.0%

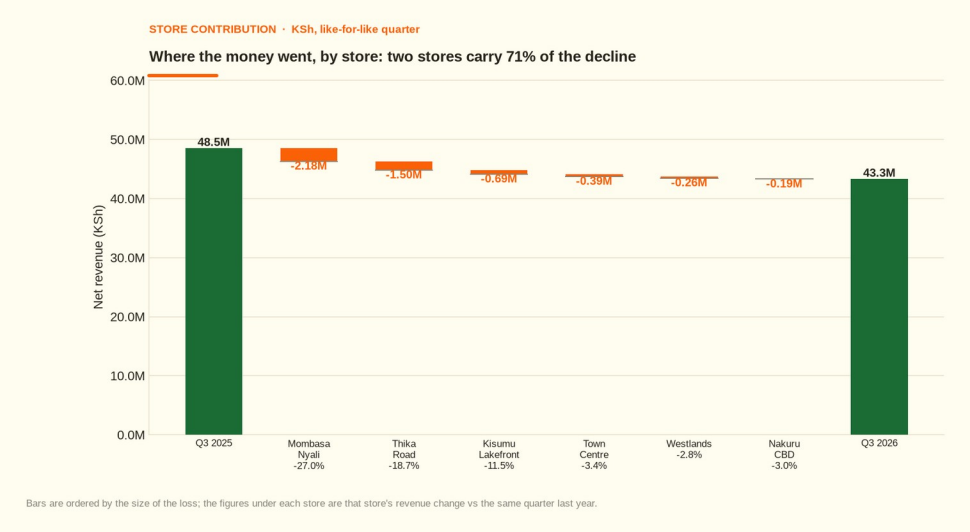
By customer segment

Rank	Slice	Revenue change	YoY	Share of group decline
1	Loyal	KSh -3.03M	-15.2%	58.2%
2	Regular	KSh -2.52M	-14.0%	48.4%

Rank	Slice	Revenue change	YoY	Share of group decline
3	At-Risk	KSh -0.85M	-14.2%	16.3%
4	Hibernating	KSh -0.04M	-2.9%	0.8%
5	New	KSh +1.28M	+40.7%	-24.5%

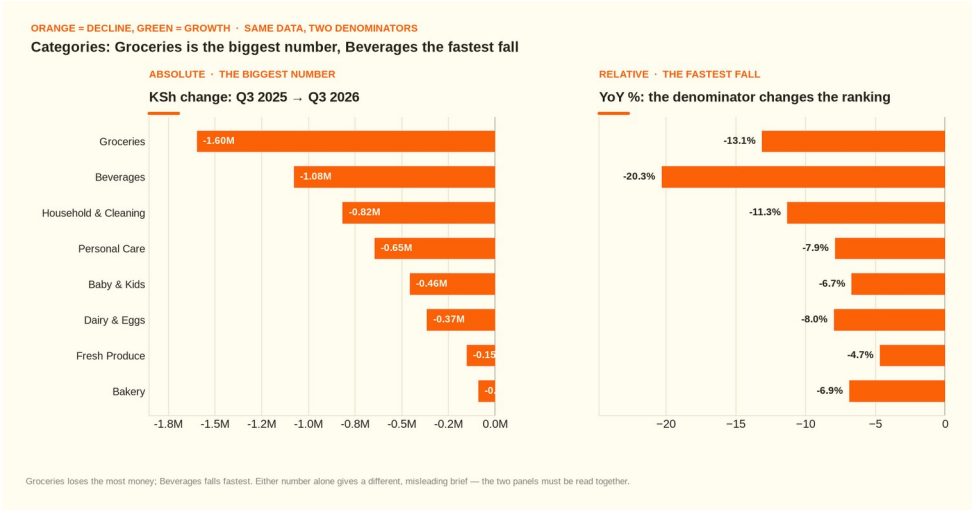
By channel

Rank	Slice	Revenue change	YoY	Share of group decline
1	In-store	KSh -4.54M	-10.8%	87.1%
2	Online	KSh -0.67M	-9.9%	12.9%



Two of six stores carry 71% of the decline.

Figure 3 — where the money went, by store.

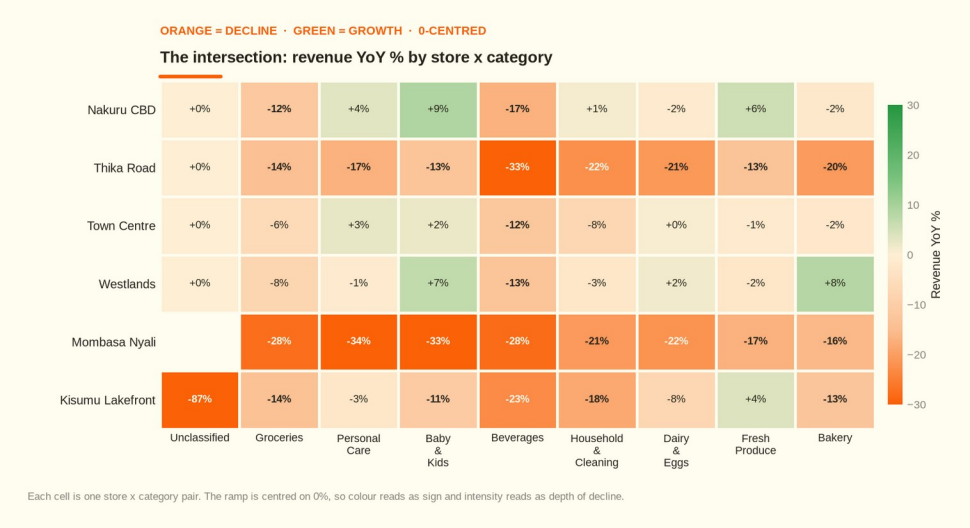


Groceries is the biggest absolute loss, Beverages the fastest fall. Either alone is a misleading brief.

Figure 4 — categories, in both KSh and %.

SECTION 7

The intersection — which store x product x segment



The decline is concentrated, not diffuse — which is why the fix can be targeted too.

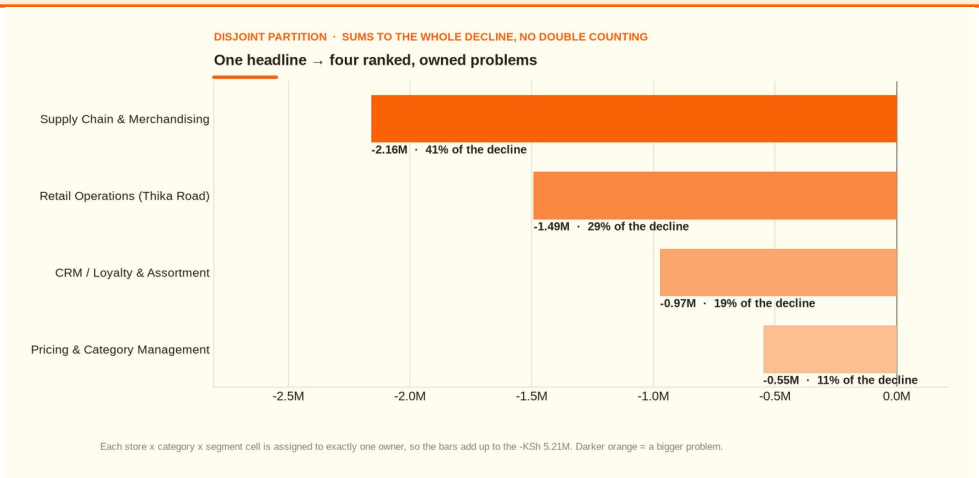
Figure 5 — revenue YoY % by store x category.

The worst 8 cells (store x category x segment) account for 23.5% of the total decline.

#	Cell	Revenue change	YoY	% of group decline
1	Mombasa Nyali | Groceries | Regular	KSh -0.28M	-33.6%	5.4%
2	Mombasa Nyali | Groceries | Loyal	KSh -0.23M	-29.6%	4.4%
3	Mombasa Nyali | Personal Care | Regular	KSh -0.22M	-41.1%	4.2%
4	Mombasa Nyali | Baby & Kids | Loyal	KSh -0.21M	-42.0%	4.0%
5	Mombasa Nyali | Personal Care | Loyal	KSh -0.19M	-35.9%	3.6%
6	Mombasa Nyali | Baby & Kids | Regular	KSh -0.17M	-34.8%	3.3%
7	Thika Road | Personal Care | Loyal	KSh -0.17M	-30.9%	3.3%
8	Mombasa Nyali | Household & Cleaning | Regular	KSh -0.17M	-37.0%	3.2%

The counter-intuitive result

The single largest cell is Mombasa Nyali | Groceries | Regular — not the category that was out of stock. At Mombasa Nyali, every category fell together because the stock failure removed the reason for the trip and the whole basket shrank with it. Measuring only the affected category would have understated this event by roughly 4.8x.



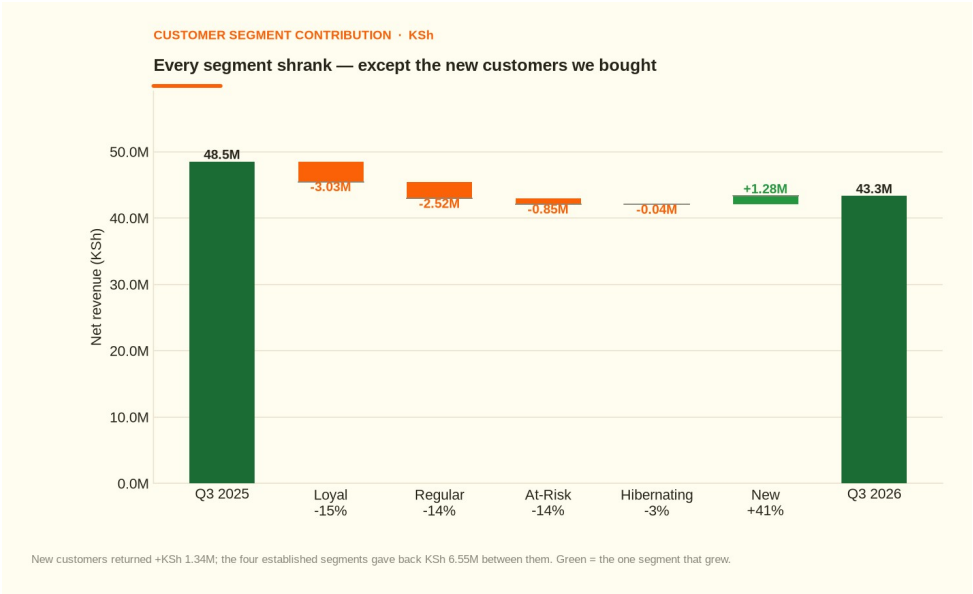
Each cell is assigned to exactly one owner, so there is no double counting.

Figure 6 — four ranked, owned problems (disjoint; sums to the whole decline).

SECTION 8

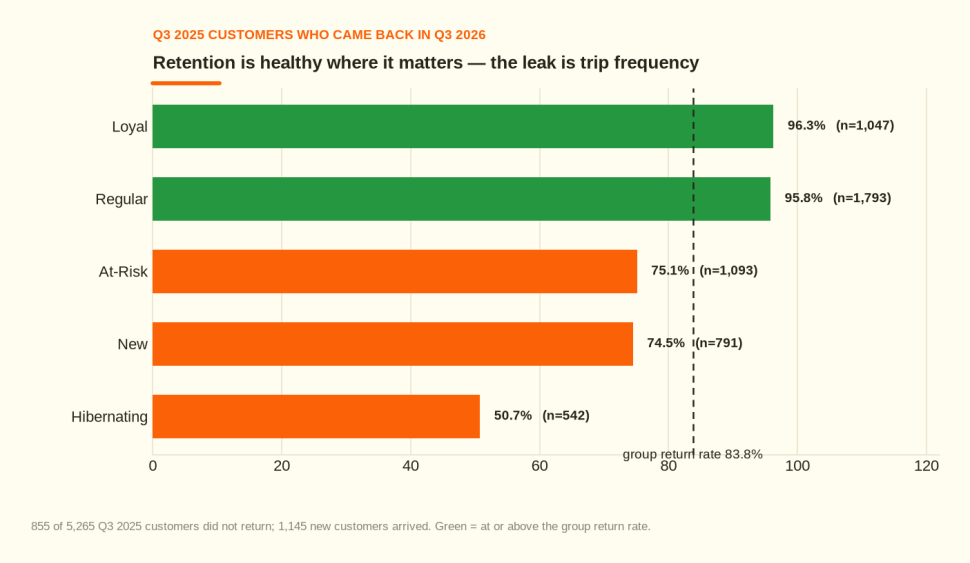
Segments and retention

Cohort metric	Value
Q3 2025 customers	5,265
Returned in Q3 2026	4,410 (83.8%)
Lapsed	855
New customers acquired	1,145



Buying growth is not a strategy while the existing base contracts.

Figure 7 — every established segment shrank except the customers we bought.



Retention is healthy where it matters. The leak is trip frequency, not churn.

Figure 8 — return rate by base-quarter segment.

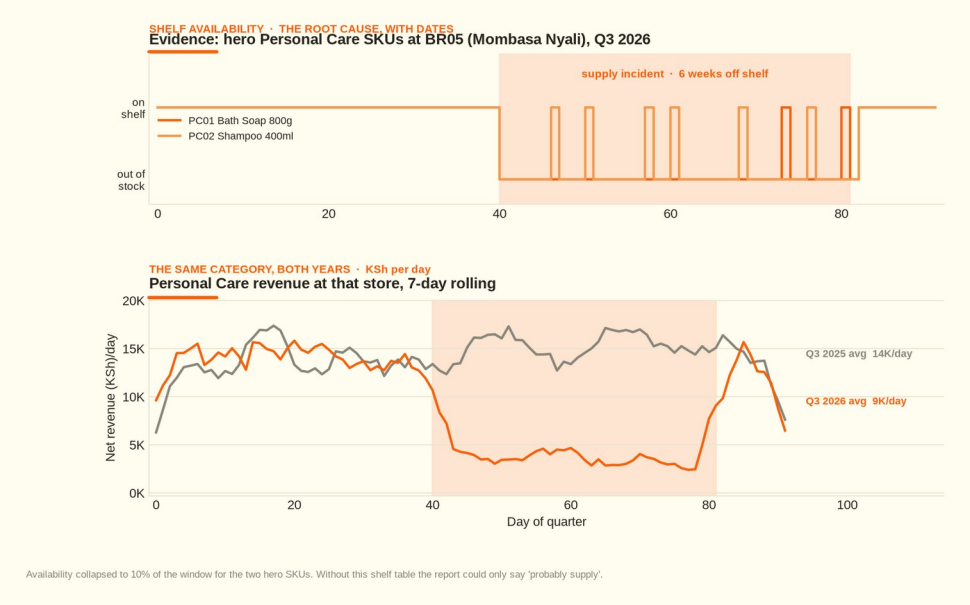
SECTION 9

The supply event, with dates

	Value
Store	Mombasa Nyali
Hero SKUs	Bath Soap 800g
Shelf availability in the incident window	9.5%
Days out of stock	76
Total cost attributed	KSh 2.16M
Of which the affected category	KSh 0.45M
Of which basket spillover (other categories)	KSh 1.71M

Why this is evidence, not a story

The shelf-availability table records, per branch, per SKU, per day, whether the item was on the shelf. That is what turns “probably a supply problem” into a dated, store-specific, owned finding.



The incident has a start date, an end date, a store and a measurable cost.

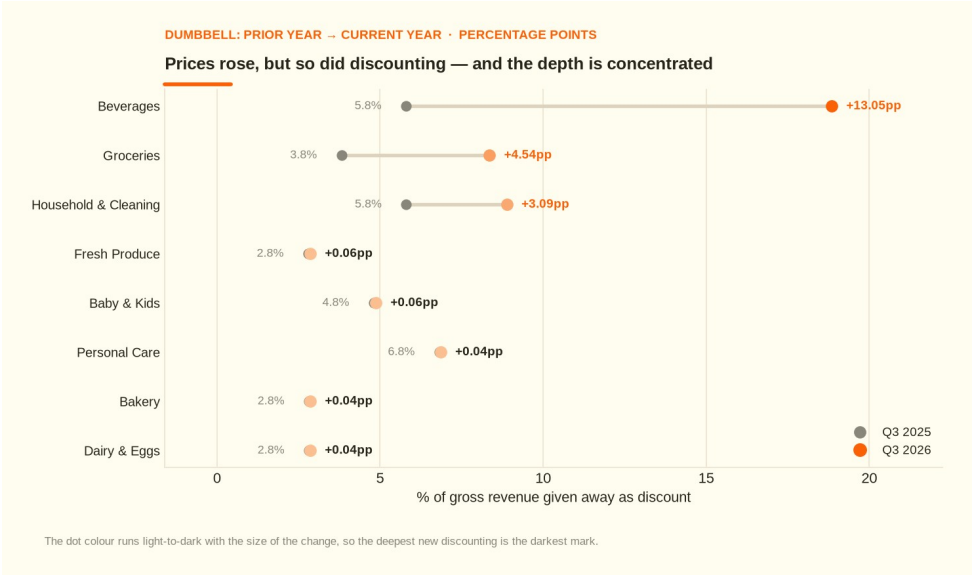
Figure 9 — shelf availability and the revenue trace that followed it.

SECTION 10

Pricing, mix and margin

	Q3 2025	Q3 2026	Change
Discount depth (% of gross)	4.80%	7.82%	+3.02pp
Gross margin	19.95%	18.57%	-1.38pp
Gross profit	KSh 9.68M	KSh 8.04M	-16.9%

Roughly KSh 1.43M of extra discount was given away this quarter, concentrated in Beverages (+13.05pp). Gross profit fell faster than revenue, so margin — not just revenue — is the thing at risk.



Prices rose, but so did discounting. The depth is concentrated.

Figure 10 — discount depth, prior year vs current year, by category.

SECTION 11

Facts, hypotheses, and the edge of what we know

A senior analysis is judged as much by what it refuses to claim as by what it concludes. Every statement below is tagged.

Statement	Tier	Note
Revenue fell 10.7%; gross profit fell 16.9%	Fact	measured from the cleaned star schema
Two stores explain 71% of the decline	Fact	measured; contribution shares sum to the group change
Hero SKUs were 9.5% available for 76 days at Mombasa Nyali	Fact	from the branch x SKU x day availability table
Active customers rose +5.5% while frequency fell -11.4%	Fact	the mechanism identity, order-independent attribution
The stock-out *caused* the basket erosion at that store	Hypothesis	the two co-move exactly in time; no causal test was run
Single-sourcing or no safety stock caused the stock-out	Hypothesis	no procurement, supplier or lead-time data in this extract
Thika Road's decline is due to the road-works	Hypothesis	the frequency-only signature is consistent with an access problem; no other local factor is recorded
Kisumu Lakefront's -11.5%	Unexplained	flagged, deliberately not back-filled with a story

Statement	Tier	Note
855 Q3 2025 customers did not return	Fact	but 83.8% did — churn is not the problem

SECTION 12

Predictive — the run-rate if nothing changes

A run-rate, not a forecast

If this quarter's frequency and basket levels persist, the annualised revenue impact is roughly KSh -20.83M (KSh 5.21M x 4). This assumes the Mombasa Nyali incident does not repeat — it should be fixed — and it ignores Q4 seasonality, which is typically stronger. Treat it as the size of the hole, not the shape of the year.

The honest reading: the quarter's damage is stable and repeatable in nature, so the cost of inaction is not zero. It is the run-rate above, minus whatever the actions in Section 13 recover.

SECTION 13

Prescriptive — what to do

Every action carries the four things an action needs to be real: an owner, an expected impact, an evidence link and a review date. Impacts are deliberately conservative and priced in gross profit — they assume partial capture.

#	Action	Owner	Why (evidence)	Expected impact	Review
1	Rebuild safety stock on the two hero Personal Care SKUs and qualify a second supplier	Supply Chain	Availability was 9.5% for 76 days; the store lost KSh 2.18M	KSh 1.51M – KSh 2.16M	2026-11-04 (+30d)
2	Rebuild the Mombasa basket: substitutes on shelf, Personal Care adjacent to Groceries, a substitution script for staff	Merchandising	AOV fell -22.6% while the category alone explains only KSh 0.45M of the KSh 2.16M damage	KSh 0.18M	2026-11-19 (+45d)
3	Thika Road access recovery: landlord and county engagement, extended hours, local marketing to restore trips	Retail Operations	Frequency fell -23.7% while basket value held at -0.3% — a pure traffic problem worth KSh 1.49M	KSh 0.89M	2026-12-04 (+60d)
4	Cap beverage promotional depth back toward 4.8%	Pricing & Category Management	Beverage discount depth rose +13.05pp; category	KSh 0.27M	2026-10-26 (+21d)

#	Action	Owner	Why (evidence)	Expected impact	Review
	of gross and require an approval gate above it		revenue fell -20.3%		
5	Convert the frequency leak: targeted win-back for the 855 lapsed customers and a second-purchase nudge for the 1,145 new ones	CRM / Loyalty	-11.4% trips per customer across all four established segments; residual pool worth KSh 0.97M	KSh 0.49M	2027-01-03 (+90d)
6	Capture a customer identifier on every transaction; start logging supplier promised-vs-actual lead times	Data / POS & Supply Chain	0.76% of revenue is currently unattributable and the root cause of the stock-out is unanswerable	No direct revenue; removes both blind spots	2027-01-03 (next quarter)

Sequencing and dependencies

- Days 0-30: actions 1 and 4. Both are cheap and reversible, and both stop active bleeding. Action 4 is a policy change; action 1 is a purchase order.
- Days 30-60: actions 2 and 3. These need time to show up in traffic data.
- Days 60-90: action 5, which depends on knowing the base is stable first.
- Continuous: action 6. It is the only item that makes the *next* diagnosis cheaper — and this entire report exists because two data gaps had to be worked around.

SECTION 14

How we will know whether it worked

Track the same five numbers this analysis used, on the same grain, every month. An analysis that never checks its own predictions never improves.

Metric	Baseline	Target by Q1 FY2027	Watched by
Orders per customer	5.44	back above 5.44	CRM
Hero SKU shelf availability (Mombasa)	9.5%	>99%	Supply Chain
Thika Road visits per customer	-23.7% YoY	>0% YoY	Retail Ops
Beverage discount depth	7.82% of gross	<= 4.80%	Pricing
Gross margin	18.57%	back above 19.95%	Finance

APPENDIX A

Method, provenance and reproducibility

The whole project is a pipeline, not a set of notebooks. Each step consumes the previous step's outputs, so the analysis is reproducible rather than narrated.

Step	What it does
generate_data	simulate the star schema, with planted data-quality defects
clean_data	audit and repair, and write the data-quality report
run_sql	build the SQLite warehouse and run every sql/*.sql
decompose	mechanism, structure and the disjoint driver partition
five_whys	the evidential why-chain, generated from tables
write_report	the findings report and the recommendations register
make_figures	the ten decision charts, in the dashboard's visual language
write_reports	this report, the brief, the deck and the PDF
validate_against_truth	grade the diagnosis against the sealed answer key

Reproduce everything with: python3 src/run_pipeline.py (dataset seed 20260914). Note on the interpreter: the vendored numpy is built for CPython 3.12, so run the pipeline with a 3.12 interpreter that can see pylibs/ — for example make PY=/usr/bin/python3.12.

The dataset is simulated from a documented mechanism (docs/00_ground_truth.md). That is deliberate: it allows the diagnosis to be graded. outputs/validation_report.md records 14/14 falsifiable checks passing.

APPENDIX B

Index of charts

Figure	Section	The sentence it proves
One headline, three mechanisms	Mechanism	The 10.7% fall is one customer gain against two larger losses — it is not one event.
Customers up, frequency down	Mechanism	Acquisition is working; repeat behaviour is not. Different problem, different owner.
Where the money went, by store	Structure	Two stores carry 71% of the decline.
The intersection: store x category	Intersection	The decline is concentrated, not diffuse — so the fix can be too.
Categories: KSh and % side by side	Structure	Groceries is the biggest absolute loss, Beverages the fastest fall. Either alone

Figure	Section	The sentence it proves
		misleads.
Every segment shrank except the customers we bought	Segments	Buying growth is not a strategy while the existing base contracts.
Who came back	Segments	83.8% returned — the leak is trips, not churn.
The supply failure, with dates	Evidence	9.5% availability for 76 days at Mombasa Nyali.
Prices rose, discounting rose faster	Pricing	KSh 1.43M of extra discount in one quarter.
Four ranked, owned problems	Actions	Disjoint and audited: the bars sum to the whole decline with no double counting.

APPENDIX C

The evidential 5-Whys

Every “why” is answered with a number and a source, and the chain stops where the data stops rather than speculating.

Why 1 — Why did group revenue fall?

Net revenue fell KSh 5.17M (10.66%), from KSh 48.49M to KSh 43.32M. Volumes, not customers: active customers ROSE +5.51% while orders fell 6.48%.

Source: mechanism_by_period.csv · No — this is the question, not an answer

Why 2 — Customers grew but orders fell. Why?

Orders per customer fell -11.36% (5.44 -> 4.82 trips), and average order value fell -4.48% (KSh 1,694 -> KSh 1,619). Log-mean attribution puts KSh 2.46M of growth on new customers, KSh 5.53M against frequency and KSh 2.10M against basket value.

Source: mechanism_by_period.csv, decomp_mechanism_waterfall.csv · Partly — it names the mechanism (repeat visits, not acquisition)

Why 3 — Where is the frequency and basket loss concentrated?

Two of six stores explain 71.3% of the entire group decline: Mombasa Nyali (KSh -2.18M, -27.0%) and Thika Road (KSh -1.50M, -18.7%). Frequency fell -11.7% at Mombasa and -23.7% at Thika Road, versus -11.4% for the group.

Source: structure_by_branch.csv, structure_branch_mechanism.csv · Yes for the stores — but not yet for the cause

Why 4 — Why did Mombasa Nyali lose so much more than the rest of the chain?

Its two hero Personal Care SKUs were off the shelf for six weeks: 76 out-of-stock days out of 84 observations (9.5% shelf availability, first gap 2026-08-10). Personal Care revenue at that store fell -34.2% — but Mombasa's average order value also fell -22.6% (KSh 1,724 -> KSh 1,334) against a group AOV change of -4.5%. The missing items cost the whole basket, not just the category.

Source: fact_stockouts, stockout_incident_vs_rest.csv, structure_category_mechanism.csv · Yes — this is a supply-chain event with a date and a store

Why 5 — Why were the SKUs unavailable for six weeks?

The data shows WHAT happened (a 40-day gap on the worst single SKU, longest consecutive outage 33 days) but cannot show WHY. Replenishment, supplier and lead-time data are not in this dataset. Candidate causes — single-

sourced supply, no safety stock, a mis-set reorder point — are HYPOTHESES and must be labelled as such until procurement data is pulled.

Source: fact_stockouts (WHY is absent by design) · Stop here: the actionable unit is 'restore stock and dual-source', not 'explain the universe'